

2026 Hackathon: Syngenta Use Case Briefs

Context: what Syngenta does and where these four use cases sit

Syngenta is one of the world's largest agricultural companies. It sells two main things to farmers: crop protection products such as herbicides, fungicides, and insecticides, and seeds. **This hackathon sits entirely on the seeds side of the business, specifically vegetable seeds.**

Seed is a living product with a very long lead time. Syngenta breeds a variety, contracts growers worldwide to multiply it in the field, harvests the result, then processes the harvest into a packaged product a customer can plant. From the decision to grow to the moment a customer takes delivery can be up to 18 months, and much of the coordination still happens in Excel.

Conditioning is the first processing stage after harvest: cleaning, sizing and grading raw seed so it is fit to be treated, packed and shipped. It runs on dedicated lines with finite capacity, and switching between varieties or seed sizes costs line time that cannot be recovered.

That is why the order in which batches run matters. Sequence well, and the right customer orders ship on time while the line keeps moving. Sequence badly and the plant looks busy but still ships late. Today that decision is made by one person, by hand, reacting to whatever arrives. **Use Case 1** is about giving that person a recommendation they can trust and explain.

The other three use cases sit in different parts of the same company. **Use Case 2** is commercial. Before any seed is grown, sales and marketing have to estimate how much of each variety a market will buy and how much of that market Syngenta already holds. Those numbers drive what gets planted more than a year ahead, and today they are keyed into spreadsheets and a homegrown app with no feedback while you type.

Use Case 3 is the plumbing underneath the commercial side: the customer, territory and org-structure records that every report depends on. When a customer has no territory owner, or two systems disagree about the hierarchy, the errors surface later as arguments about whose numbers are right.

Use Case 4 is research. Long before a variety reaches a farmer, breeders run field trials and lab tests across many candidate lines and decide which ones advance. That decision data is scattered across spreadsheets, PDFs, and chat messages, and the breeding cycle currently runs at roughly twice its ideal length because of it.

Use Case 1: Plant Capacity Utilization (Conditioning scope)

Primary persona: The person who schedules the conditioning line at a seed processing plant, the one who decides the run order for incoming batches today. The job to be done: when a new batch arrives or a test result lands, they want an updated run order with a stated reason for each position, so they can keep the line running without re-planning by hand.

Success criteria: A working prototype that takes a batch list, plant capacity and open customer orders, and returns a ranked conditioning schedule with a stated reason for each position. Demo-ready means the team can inject a surprise rush batch or a failed quality test live, and the plan re-sequences and explains in plain language what changed and why.

Data: Operational extracts from the plant's scheduling and ERP systems (conditioning logs, packaging logs, pass/fail test results and per-line schedules), supplemented as needed with batch records (species, variety, quantity, seed size, arrival date), processing steps and durations, changeover rules grouped by variety, and a list of open customer orders.

Out of scope: The treatment and packing stage that runs after conditioning, any second facility, crop planning, routing work between plants, sensor-level plant floor data, and predicting when harvested batches will arrive at the plant.

Non-negotiable constraints: No connection to live production systems; teams work from extracts or synthetic data. The agent recommends and a human validates before any plan is treated as final. Frame the output as a recommendation with an explanation, not a mathematically optimal schedule. Judges should not be comparing solver quality.

Problem statement: The scheduler on a seed conditioning line sequences batches by reacting to whatever shows up, with no visibility into which batch is holding up a customer order and no record of why the plan changed. Build an agent that produces a prioritized run order, re-sequences automatically when a rush batch or a failed test lands, and explains each change in plain language.

What teams could build (but not limited to):

- A scheduling agent that ranks batches with a stated reason for each position
- A re-planning loop that re-sequences and summarizes impact when a rush lands
- An "explain my batch" chat for sales: why it's waiting, when it ships, what would move it up

Nice to have: The plant sponsor is available async during hack week for questions, and the target screen concept they described can be shared as a reference sketch.

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Use Case 2: Market Intelligence and Demand Capture (MVP scope)

Primary persona: A sales rep or product specialist who submits a monthly demand number for their territory. The job to be done: when they enter a demand figure, they want to see straight away what it does to market share and to the year-to-go position, so they can submit a number they can defend rather than a guess.

Success criteria: A working tool where the rep enters a demand number for one locked crop and geography and immediately sees market-share impact, year-to-go impact, an implausibility flag against history, and a one-sentence justification field that becomes structured reasoning. Demo-ready means live entry produces a live recalculation with no next-day lag, which is the core complaint from the business today.

Data: An anonymized one-country sample (market-sizing data plus matching grower-potential data from the CRM) and anonymized historical demand spreadsheets, supplemented as needed with a crop, country and state-level hierarchy, hectares, plant density, quantity and price fields, and a monthly demand table.

Out of scope: Any forecasting model, the sales-territory and geographic-rollup layer, scraping anything beyond public agricultural sources, and any crop or geography beyond the one locked for the demo.

Non-negotiable constraints: Sales figures are synthetic unless explicitly cleared. Web research is limited to public agricultural sources such as national statistics agencies. No live access to production systems.

Problem statement: Sales reps update demand forecasts in disconnected spreadsheets and a homegrown app with no real-time feedback, so a number cannot be validated or defended while the conversation is still happening. Build a tool, scoped to one crop and one geography, that shows live market-share and year-to-go impact as a number is entered, flags implausible entries, and captures the justification as structured history.

What teams could build (but not limited to):

- A demand-capture interface with real-time market-share and YTG feedback
- An input-validation copilot that flags implausible entries against history
- A consensus assistant that drafts the reasons-to-believe narrative (stretch)

Nice to have: A hand-drawn UI sketch from the process owner can be shared as a literal wireframe reference.

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Use Case 3: Commercial Data Integrity Agent (provisional)

Primary persona: A commercial data steward. The job to be done: when customer records are loaded or changed, the steward wants to know which records break which rule and what the fix is, so they can clean the data before it distorts reporting.

Success criteria: A working agent that scans a customer master export against a defined set of rules, flags each violation with its business impact, and recommends an owner and a fix.

Out of scope: Access provisioning and identity management. Auto-writing to master data. Attempting both hierarchy reconciliation and data quality in one deliverable; teams pick one.

Non-negotiable constraints: The agent recommends and routes; it never writes to production master data.

Problem statement: Commercial data stewards manually search customer records for duplicates, missing attributes, and orphaned accounts, coordinating fixes by email with no way to confirm a record is even correct. Build an agent that scans a CRM export against defined rules, flags violations by business impact, and recommends an owner and a fix.

What teams could build (but not limited to):

- A rule-driven data-quality agent over the customer master
- Business-impact scoring and owner routing
- A natural-language rule-authoring interface

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Use Case 4: R&D Data Source Unification (MCP)

Primary persona: A senior plant breeder, typically with 30 or more years of experience, a non-technical background, and a working day spent mostly in spreadsheets and on paper. The job to be done: when deciding which candidate lines advance, they want the trial, lab and operational data in one place with a recommendation and its reasoning, so they can make the call without chasing five systems.

Success criteria: A working assistant that ingests the four mock data sources (trial, operations, lab/pedigree, germplasm), answers breeder questions in natural language, and returns a red/amber/green recommendation per candidate with a stated reason. Demo-ready means: show a flagged candidate with a one-line explanation the breeder can challenge and override, with the override logged.

Data: Anonymized mock tables with real column headers across the four sources, plus the pass/fail scoring logic.

Out of scope: Any connection to a live research system.

Non-negotiable constraints: A human in the loop is mandatory, not optional, because an earlier pilot left breeders wary of automated decisions. Every recommendation must show its reasoning, never just a verdict. Frame it as an assistant, not a replacement.

Problem statement: Plant breeders track trial, lab, and operational data across disconnected Excel sheets, PDFs, and WhatsApp messages, re-entering the same information multiple times, roughly doubling the breeding cycle length. Build an assistant that unifies the mock sources behind a natural-language interface and triages candidates red/amber/green with an explanation, always deferring the final call to the breeder.

What teams could build (but not limited to):

- A data-unification (MCP) layer over the mock sources
- Natural-language query interface for breeders
- Red-amber-green triage with explained recommendation and override/feedback loop

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